

ADDICTION ANALYSIS REPORT

Prepared By: Oreva Egwebe

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Section 1 — Data Understanding & Exploration

1.1 Dataset Overview:

This analysis is based on the **Addiction Dataset**, which contains **3,000 observations** and **25 variables** representing a diverse global demographic (ages 15–79). The report investigates behavioral patterns, health impact, and demographic vulnerabilities related to smoking and alcohol addictions.

- **Data Types:** The dataset is a mix of **numerical** (`age`, `annual_income_usd`, `smokes_per_day`, `drinks_per_week`), **categorical** (`gender`, `education_level`, `mental_health_status`, `exercise_frequency`), and **boolean** (`has_health_issues`) variables.
- **Missing Values:** Significant missing values were identified in `education_level` (420), `social_support` (753), and `therapy_history` (1,014). To maintain dataset integrity, these missing categorical values were replaced with **"unavailable"**.
- **Inconsistent Values:** There were some instances where the starting age for drinking or smoking was greater than the current age, these records were identified and replaced with their current age.
- **Standardizing the smoking metric** involved converting `smokes_per_day` to `smokes_per_week` (multiplied by 7) to allow for a direct comparison with the weekly drinking metric.

1.2 Initial Observations:

- **Early Onset:** The minimum age for starting smoking or drinking is **10**, while the average age is **24**. This shows that they likely face higher long-term addiction risks.
- **Unrealistic BMI Values:** The minimum BMI was 9.1, which is far below the healthy range of 18.5. This is a strong indicator of data entry error or an outlier. Such values were replaced with the mean BMI of the dataset to reduce their impact.
- **Extreme Income Values:** The annual income ranged from \$560 to \$199,951. The high standard deviation and the presence of students and unemployed individuals with incomes over \$100k suggest either highly affluent demographics or data reporting inconsistencies.
- **Logical Inconsistencies in Age:** There were **462** records where the age at which a person started smoking was greater than their current age, which is logically impossible.

Similarly, **505** records had a similar issue for drinking. These were corrected by setting the starting age equal to the current age.

1.3 Data Quality Assessment

To validate the data, the following checks were performed:

- **BMI Validation:** Checked for BMI values below 15 and replaced them with the mean BMI to account for outliers or data errors.
- **Age Validation:** Checked for records where `age_started_smoking` or `age_started_drinking` exceeded the current age.
- **Income Validation:** Checked `annual_income_usd` for individuals with an employment status of "Unemployed" or "Student" to identify unusual high-income values or data entry errors.
- **Health Status Validation:** Checked for individuals with a high BMI (>40) who were flagged as not having health issues to identify potential high-risk cases for further review.

Section 2 — Exploratory Data Analysis

2.1. Demographic Analysis

- **Age Group:** The 18-25 and 65+ age groups exhibit the highest average consumption for both smoking and drinking. This suggests that younger adults and the elderly are at high-risk for substance abuse.
- **Gender:** Females have higher average smoking habits than males. Interestingly, average drinking habits are similar across all genders.
- **Education Level:** High school graduates have the highest smoking rates, whereas college and postgraduate graduates show higher drinking rates with only moderate smoking levels. This points to potential social and cultural factors at play, where higher education might correlate with less smoking but not less drinking.
- **Employment Status:** Self-employed individuals have the highest average smoking and drinking habits. This could be attributed to the stress, irregular hours, or social norms associated with self-employment.

2.2 Geographical Analysis

Geographical analysis highlights significant regional disparities in substance use.

- **Countries:** Korea, Congo, and Saudi Arabia consistently appear among the top countries with the highest total consumption of both alcohol and cigarettes.
- **Cities:** Port Jennifer stands out as the city with the highest total alcohol and cigarette consumption. Port James also appears in the top rankings for both substances.

The high substance abuse rate in countries and cities could be explained by a combination of factors, including cultural norms, availability of substances, pricing and taxation policies, public health interventions, and socioeconomic conditions.

2.3 Income and Addiction

The analysis shows a notable relationship between income and substance use:

- **Smoking:** Individuals earning less than \$30,000 annually exhibit the highest smoking habits. This trend suggests that lower-income groups may face more stress, have fewer coping mechanisms, or be more susceptible to targeted marketing.
- **Drinking:** While the <\$30k group also has the second-highest drinking habits, the 120k-150k income group shows the lowest drinking habits.

Section 3 — Behavioral Analysis

The analysis reveals strong links between lifestyle choices and substance abuse:

3.1 Diet and Exercise: The analysis shows that diet quality and exercise frequency has little(constant) effect among smoking individuals, while individuals with poor diet quality tend to drink more. Also, people who exercise weekly tend to drink less.

3.2 Sleep: A regression plot indicates that there is no correlation between sleep hours and the number of cigarettes smoked and alcohol consumed per week.

3.3 Addiction Onset

- **Starting Age:** The analysis shows 10 years as the starting age for both smoking and drink habits
- **Quitting Attempts:** A scatter plot between attempts to quit and smokes per day shows no relationship which may be due to enough granular data.

3.4 Mental Health Relationship: The analysis showed no meaningful difference in smoking intensity between mental health groups. Drinking patterns show that persons with "Good" mental health tend to drink more which is logically conflicting and needs to be further investigated.

Section 4 — Health Impact Analysis

4.1 BMI and Lifestyle: The analysis of BMI and substance use does not show a strong correlation. This suggests that while BMI alone may not be a perfect predictor of substance use, it is part of a broader picture of health behaviors.

4.2 Sleep Analysis: The analysis between sleep hours and substance abuse shows a horizontal distribution of points, indicating no relationship between sleep hours and the number of cigarettes smoked or alcohol consumed per week.

Section 5 — Communication and Insights

5.1 Key Insights

1. **Early Onset of Substance Use:** The substance use at a young age emphasizes the need for early prevention programs.
2. Addiction is prevalent among self-employed individuals and individuals earning less than \$30,000 in annual income. However, annual income has no strong relationship with substance abuse.
3. Poor diet and lack of exercise are associated with higher rates of drinking, suggesting a pattern of health-risk behaviors.
4. Females have a higher smoking rate than other genders but drinking rates are constant across all genders.
5. Certain countries (Korea, Congo and Saudi Arabia) and cities (Port Jennifer) have significantly high rates of substance use, indicating a need for culturally targeted public health interventions.

5.2 Policy Recommendations

1. **Targeted Early Interventions:** Implement school-based education and mental health support for children aged 10-15 to educate them on the dangers of substance abuse and also prevent the onset of addiction.
2. **Support for High-Risk Demographics:** Develop tailored intervention programs for high-risk groups such as the self-employed, high school graduates, and lower-income individuals. This could include stress management workshops, accessible mental health support, and resources for quitting.
3. **Integrate Mental Health Services:** Healthcare providers should screen for substance abuse during mental health consultations and provide adequate mental health support in addiction recovery programs.
4. **Promote Healthy Lifestyles:** Launch public health campaigns that highlight the importance of diet, exercise, and healthy living. Encourage physical activity and healthy eating habits as part of a comprehensive approach to wellness and addiction prevention.
5. **Geographically Targeted Campaigns:** Initiate public health awareness campaigns to the regions with the highest consumption rates, such as Korea - to understand and address local contributing factors.